

Lec 4-2. (Sun, Sep 28, 2025)

- Monotonicity of functions (4.3)
 - Intervals of monotonicity
 - First derivative test
- Concavity (4.4)

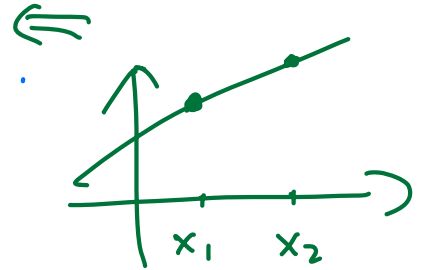
Monotonicity

Def: Let f be a function defined on an interval I ,

and let x_1 and x_2 be any two points in I .

1. If $f(x_2) > f(x_1)$ whenever $x_1 < x_2$,

then f is said to be increasing on I .

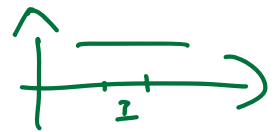


2. If $f(x_2) < f(x_1)$ whenever $x_1 < x_2$,

then f is said to be decreasing on I .

3. If $f(x_2) \geq f(x_1)$ whenever $x_1 < x_2$,

then f is said to be nondecreasing / weakly increasing on I .



4. If $f(x_2) \leq f(x_1)$ whenever $x_1 < x_2$,

then f is said to be nonincreasing / weakly decreasing on I .

Def. A function that is increasing or decreasing on I

is said to be monotone / monotonic on I .

Cor 4.3.3. Suppose f is cont' on $[a, b]$ and diff' on (a, b) .

$$a \leq x_1 < x_2 \leq b \Rightarrow \exists c \in (x_1, x_2),$$

1. If $f'(x) > 0$ at each point $x \in (a, b)$,
then f is increasing on $[a, b]$.

$$f'(c) = \frac{f(x_2) - f(x_1)}{\underbrace{x_2 - x_1}_{> 0}} > 0$$

2. If $f'(x) < 0$ at each point $x \in (a, b)$,
then f is decreasing on $[a, b]$.

$$\Rightarrow f(x_2) - f(x_1) > 0.$$

Pf: Prove by contradiction and use MVT!

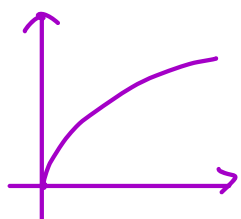
Rmk. The statements still hold if (a, b) and $[a, b]$

are replaced by (a, ∞) and $[a, \infty)$ or $(-\infty, b)$ and $(-\infty, b]$ respectively.

e.g. 4. Prove that $f(x) = \sqrt{x}$ is increasing on $[0, \infty)$.

Pf: As f is cont' on $[0, \infty)$ and diff' on $(0, \infty)$ and

$$f'(x) = \frac{1}{2} x^{-\frac{1}{2}} > 0, \quad \forall x \in (0, \infty),$$



we have f is increasing on $[0, \infty)$ based on

the extended version of Cor 4.3.3.

□

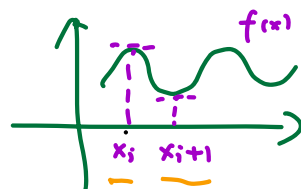
Intervals of Monotonicity

Rmk.

1. Suppose that $x_1, x_2, \dots, x_n$ are all critical points of a function f

with $x_1 < x_2 < \dots < x_n$.

• Let x_i and x_{i+1} be two consecutive critical points.



• If f' is continuous on $[x_i, x_{i+1}]$, then

either $f'(x) > 0$, $\forall x \in (x_i, x_{i+1})$,

$\exists \bar{x} \in (x_i, x_{i+1})$, s.t.

$$f'(\bar{x}) = 0$$

or $f'(x) < 0$, $\forall x \in (x_i, x_{i+1})$. [Think about why?]

• By cor 4.3.3.

① f is increasing on $[x_i, x_{i+1}]$, if $f'(c) > 0$ for some $c \in (x_i, x_{i+1})$;

② f is decreasing on $[x_i, x_{i+1}]$, if $f'(c) < 0$ for some $c \in (x_i, x_{i+1})$.

2. Similar statements can be stated for intervals $(-\infty, x_1]$ and $[x_n, \infty)$.

e.g. For the function $f(x) = x^3 - 12x - 5$, analyze the monotonicity

Sol¹: Step 1. Take derivative and solve critical points

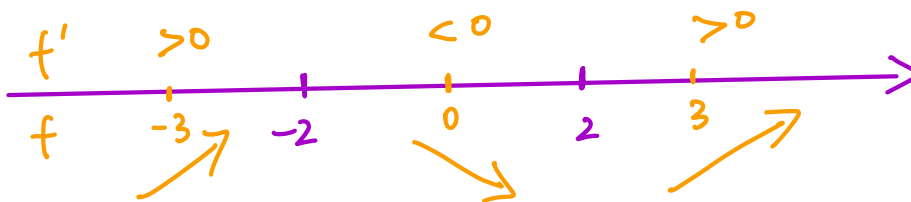
$$f'(x) = 3x^2 - 12 = 3(x-2)(x+2).$$

Let $f'(x) = 0$. We get $C_1 = 2$ and $C_2 = -2$.

Step 2. Analyze each interval.

By the remarks before, we only need to pick a sample point on each interval to test the monotonicity of f :

$$f'(-3) = 3(x-2)(x+2)|_{x=-3} > 0, \quad f'(0) = 3 \cdot (-2) \cdot 2 < 0, \quad f'(3) = 3 \cdot (3-2)(3+2) > 0$$



Since $f'(-3) > 0$, $f'(0) < 0$, $f'(3) > 0$,

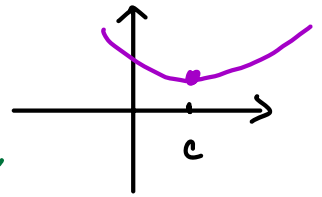
we have f is increasing on $(-\infty, -2)$,

decreasing on $(-2, 2)$,

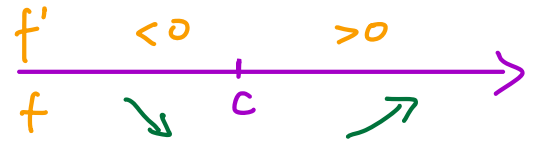
and increasing on $(2, \infty)$.

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First derivative test for local extrema



Suppose that c is a critical point of a continuous function f , and that f is differentiable at every point in some interval containing c except possibly at c itself.



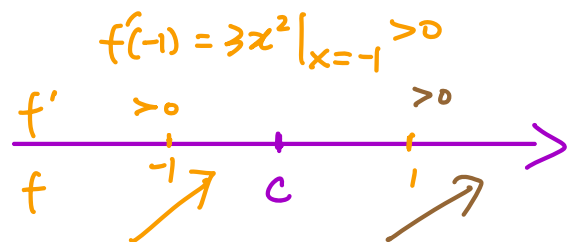
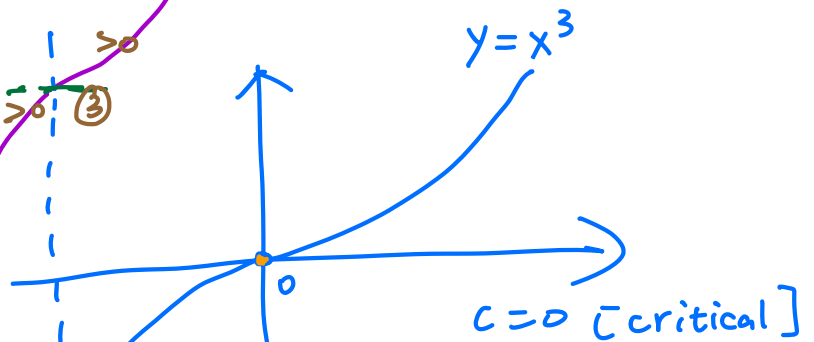
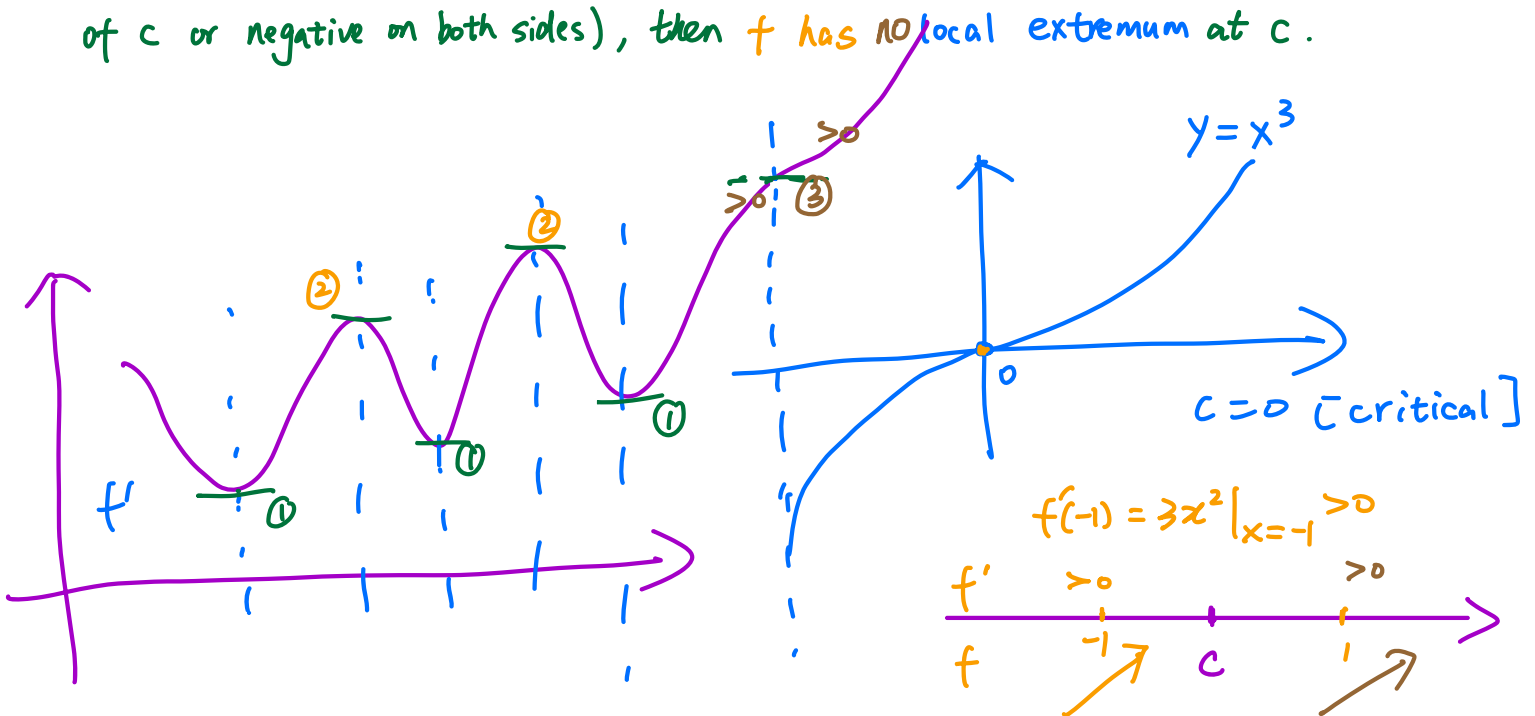
Moving across this interval from left to right,

1. if f' changes from negative to positive at c , then f has a local min at c ;

2. if f' changes from positive to negative at c , then f has a local max at c ;



3. if f' does not change sign at c (that is, f' is positive on both sides of c or negative on both sides), then f has no local extremum at c .



$$f'(-1) = 3x^2 \big|_{x=-1} > 0$$

$$f'(1) = 3x^2 \big|_{x=1} > 0$$

Rmks. For the conditions above, more formally:

1. There exists $\delta > 0$ such that $f'(x) < 0$ for all $x \in (c-\delta, c)$
and $f'(x) > 0$ for all $x \in (c, c+\delta)$;
2. There exists $\delta > 0$ such that $f'(x) > 0$ for all $x \in (c-\delta, c)$
and $f'(x) < 0$ for all $x \in (c, c+\delta)$;
3. There exists $\delta > 0$ such that $f'(x) > 0$ for all $x \in (c-\delta, c+\delta) \setminus \{c\}$
or $f'(x) < 0$ for all $x \in (c-\delta, c+\delta) \setminus \{c\}$.

Pf: 1. By Cor 4.3.3, f is decreasing on $(c-\delta, c)$ and increasing on $(c, c+\delta)$,

which means that $f(x) > f(c) \quad \forall x \in (c-\delta, c)$
and $f(x) > f(c) \quad \forall x \in (c, c+\delta)$.



Hence, $f(x) \geq f(c) \quad \forall x \in (c-\delta, c+\delta)$,

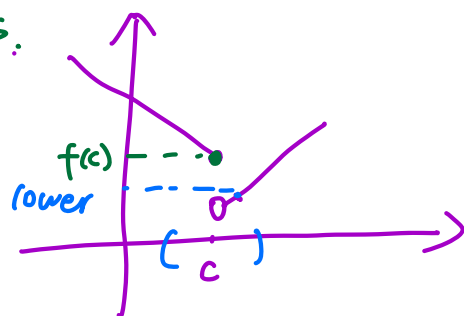
i.e. f has a local minimum at c .

□

2 and 3. Similar.

Q: Is the continuity condition essential? Yes.

- c is a critical point ($f'(c)$ DNE finitely)
- f is diff' on $(c-\delta, c+\delta) \setminus \{c\}$.
- f is not cont' and $f(c)$ is not a local min.



e.g.5. Find all absolute and local extrema for

$$f(x) = x^{4/3} - 4x^{1/3}, \quad x \in \mathbb{R}.$$

Sol: Step 1: Solve all c s.t. $f'(c) = 0$.

$$\begin{aligned} f'(x) &= \frac{4}{3} x^{1/3} - \frac{4}{3} x^{-2/3} \\ &= \frac{4}{3} x^{-2/3} (x - 1), \quad x \neq 0. \end{aligned}$$

$$\therefore f'(c) = 0 \quad \Rightarrow \quad c = 1$$

Step 2: deal with a special point.

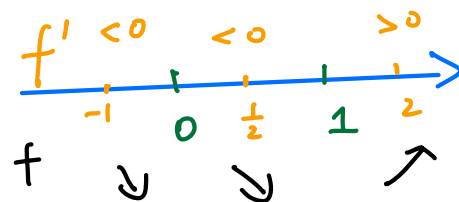
For $x=0$, since

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{x^{4/3} - 4x^{1/3}}{x} = \lim_{x \rightarrow 0} \left(x^{1/3} - 4x^{-2/3} \right) \text{ does}$$

not exist finitely.

Hence $f'(0)$ does not exist.

Hence, critical points are $x=0$ and $x=1$.



Step 3: analyze each interval and pick up a sample point.

$$\text{Since } f'(c) = \frac{4}{3} x^{-2/3} (x - 1) \Big|_{x=-1} < 0$$

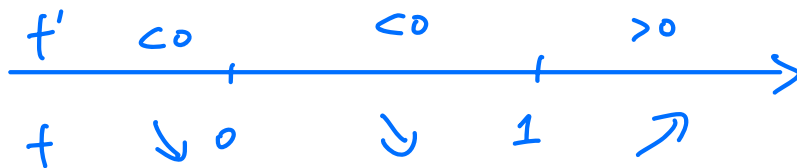
$$f'(c) = \frac{4}{3} x^{-2/3} (x - 1) \Big|_{x=1/2} < 0$$

$$f'(2) = \frac{4}{3} x^{-\frac{2}{3}} (x-1) \Big|_{x=2} > 0,$$

$\therefore$ at $x=0$, f' does not change sign and f has no extrema (case 3),

and at $x=1$, f' changes sign from negative to positive and f has a local min (case 1).

That is,



By the first derivative test, $x=1$ is the only local extremum point.

Step 4. Summarize.

f is decreasing on $(-\infty, 1)$,

and increasing on $(1, \infty)$,

and $x=1$ is the absolute min of f .

□

Q: Is it true? The function f defined by

$$f(x) = \begin{cases} x^2 (1 + \sin \frac{1}{x}) & , x \neq 0 \\ 0 & , x = 0 \end{cases}$$

has a local minimum at $x=0$.

Draft: $\forall x \neq 0, f'(x) = 2x(1 + \sin \frac{1}{x}) + x^2(\cos(\frac{1}{x}))(-\frac{1}{x^2})$

— hard to find $f'(x) = 0 \dots$

Pt: Step 1. directly analyze!

$$\forall x \neq 0, f(x) = x^2 (1 + \sin \frac{1}{x}) \geq x^2 (1 - 1) = 0.$$

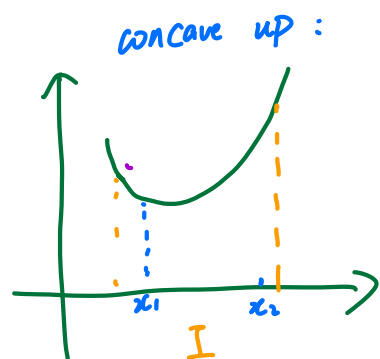
$\therefore f$ has an absolute minimum at $x=0$, and also a local minimum at $x=0$.

□.

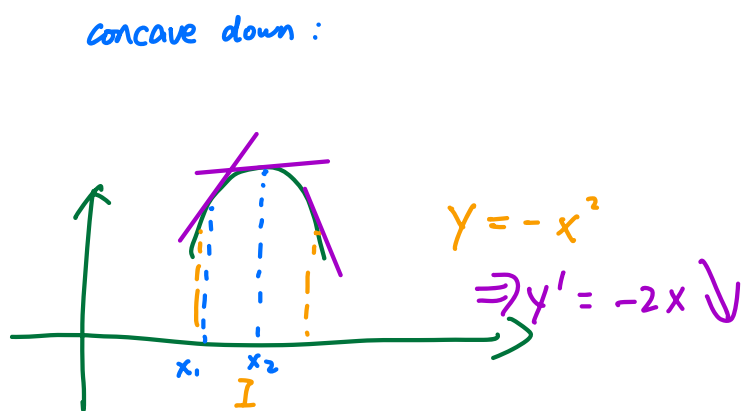
• Concavity

Def: The graph of a differentiable function $y=f(x)$ is

- (a) **concave up** on an open interval I if f' is **increasing** on I ;
- (b) **concave down** on an open interval I if f' is **decreasing** on I .



$$y = x^2 + bx + c$$
$$y' = 2x + b$$



The property of being concave up or concave down is called

[Tips: use x^2 and $-x^2$ for memory.]

Rmks

- We may say that " f is **concave up**" instead of "the graph of $y=f(x)$ is **concave up**" for simplicity.
- One may extend the definition to say that f is concave up (or down) on $[a, b)$, $(a, b]$ or $[a, b]$, with f'_+ and f'_- replacing f' for the left and right endpoints, respectively.
- In some textbooks, "**concave up**" = "**convex**".

and "concave down" = "concave".

Please use "concave up" or "concave down" in this course.

• Second derivatives and concavity

Step 1. suppose $(f')' = f''(x) > 0$, $\forall x \in (a, b)$

and f' is continuous on $[a, b]$.

Step 2. By corollary 4.3.3, f' is increasing on $[a, b]$.

Step 3. By definition, f is concave up on $[a, b]$.

A similar statement can be made to the case where $f''(x) < 0$.

Thm. Let f be a function where f' is continuous on $[a, b]$.

• If $f''(x) > 0 \quad \forall x \in (a, b)$, then f is concave up on $[a, b]$.

• If $f''(x) < 0 \quad \forall x \in (a, b)$, then f is concave down on $[a, b]$.

Pf: shown in the above yellow part.

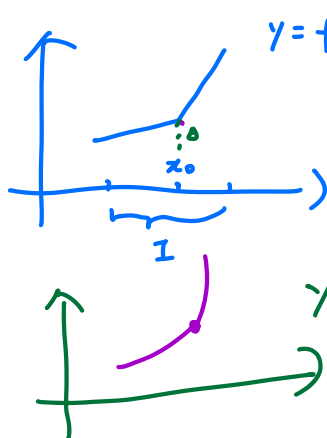
□.

Rmks :

- The theorem can be extended to cover $[a, \infty)$, $(-\infty, b]$, $(-\infty, \infty)$.
- This theorem is NOT the definition of concavity.

e.g. f' is increasing on I $\nRightarrow f'' > 0$ on I

f is concave up on I



f' is increasing on I .
But f'' is not defined at x_0 !

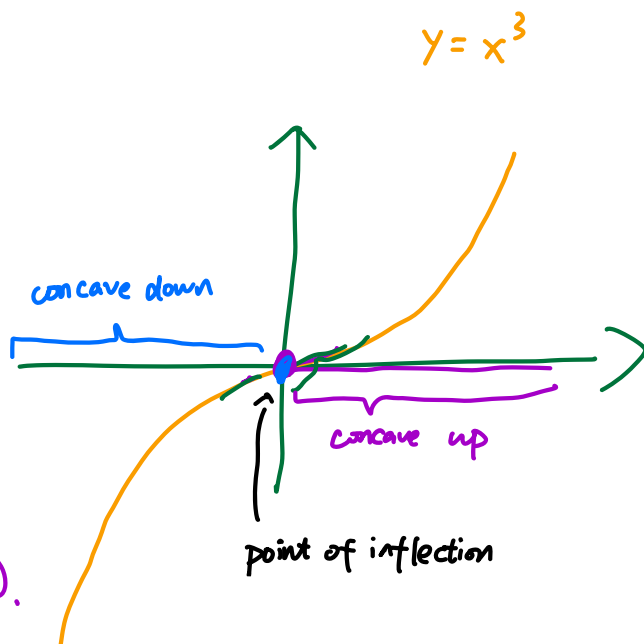
e.g. 1. Consider $y = f(x) = x^3$.

Then $f'(x) = 3x^2$,

$f''(x) = 6x$.

We have $f''(x) > 0$ for any $x \in (0, \infty)$

and $f''(x) < 0$ for any $x \in (-\infty, 0)$.



We have f is concave up on $[0, \infty)$

and f is concave down on $(-\infty, 0]$.

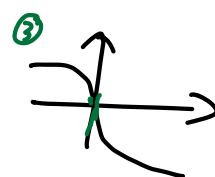
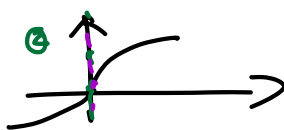
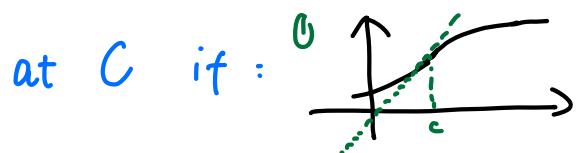

when continuous,
 f is $\uparrow$ on (a, b)
 $\Leftrightarrow f$ is $\uparrow$ on $[a, b]$

[We can also write f is concave up on $(0, \infty)$, e.g..

Because a single point at 0 does not change the monotonicity of f' (hence the concavity on the whole interval) and in this example, f'' is defined for any point in $(-\infty, \infty)$ and hence so is f' .]

$$f'(c) = L < \infty$$

Def: A function f is said to have a point of inflection (or inflection point)



• f has a tangent line (which could be vertical) at C ;

(i.e., f is diff' at C or $\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = \infty$ or $\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = -\infty$)

• $\exists \delta > 0$ such that the concavity of f on $(c - \delta, c)$ is different from the concavity of f on $(c, c + \delta)$.

Rmk . A continuous curve f is said to have a vertical tangent

at C if $\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = \infty$ or $\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = -\infty$.

e.g. ① $y = f(x) = x^{1/3}$

Step 1. check the tangent!

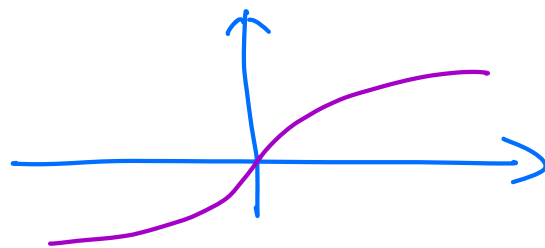
$$f'(x) = \frac{1}{3} x^{-\frac{2}{3}}, \quad x \neq 0$$

$$\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{x^{1/3} - 0}{x - 0} = \lim_{x \rightarrow 0^+} \frac{1}{x^{2/3}} = \infty.$$

$$\lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^-} \frac{x^{1/3} - 0}{x - 0} = \lim_{x \rightarrow 0^-} \frac{1}{x^{2/3}} = \infty$$

$$\therefore \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \infty \Rightarrow f \text{ has a vertical tangent.}$$

odd function ✓



Step 2. check the concavity piecewise.

$$f'(x) = \frac{1}{3} x^{-\frac{2}{3}}, \quad x \neq 0.$$

$$f''(x) = -\frac{2}{9} x^{-\frac{5}{3}}, \quad x \neq 0.$$

$$\left(\begin{array}{l} f''(x) > 0 \text{ whenever } x < 0 \\ f''(x) < 0 \text{ whenever } x > 0 \end{array} \right)$$

For $x < 0$; on the left, $f''(x) > 0$ on $(-\delta, 0)$.

$\Rightarrow f$ is concave up on $(-\delta, 0)$.

on the right, $f''(x) < 0$ on $(0, \delta)$.

$\Rightarrow f$ is concave down on $(0, \delta)$.

$\Rightarrow$ the concavity of f on $(-\delta, 0)$ is

different from the concavity of f on $(0, \delta)$

$\Rightarrow f$ has an inflection point at 0. □

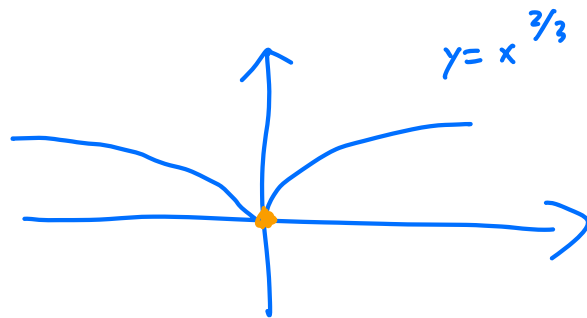
even function

e.g. ②. $y = f(x) = x^{2/3}$

Step 1: check the tangent!

$$\begin{aligned}\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h - 0} &= \lim_{h \rightarrow 0^+} \frac{h^{2/3}}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{1}{h^{1/3}} = \infty.\end{aligned}$$

$$\begin{aligned}\lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h - 0} &= \lim_{h \rightarrow 0^-} \frac{h^{2/3}}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{1}{h^{1/3}} = -\infty.\end{aligned}$$



$$\therefore \lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h - 0} \neq \lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h - 0}.$$

$\therefore f$ does not have a tangent (including vertical tangents)

at 0, so 0 is NOT an inflection point.

$$\therefore f'(x) = \frac{2}{3} x^{-1/3}, \quad x \neq 0$$

$$f''(x) = -\frac{2}{9} x^{-4/3}, \quad x \neq 0$$

$$\therefore f'' < 0 \text{ on } (-\infty, 0),$$

$$f'' < 0 \text{ on } (0, \infty).$$

$\therefore \forall x_0 \neq 0$, x_0 cannot be an inflection point.

$\therefore$ There is no inflection point for f .

□.